

Dear Students: There are four total Directed Learning Activity sessions which focus on genetics content within the MSK module. This material will be discussed during two clicker sessions in the MSK module which are called

Inherited Disorders of the Musculoskeletal System and Triplet Repeat Disorders & Pharmacogenetics.

Required reading for *Triplet Repeat Disorders & Pharmacogenetics* will entail:

1. Readings from the text
2. Video describing trinucleotide repeat disorders
3. Video describing pharmacogenetics examples
4. Pdf files for the two videos

The classroom time will consist of clicker questions and discussions based on the directed learning assignments and cumulative information from the FTM module of BPM1.

Readings: From Korf and Irons *Human Genetics and Genomics*, 4th edition:

1. The following pages from the Korf and Irons *Human Genetics and Genomics* book
 - a. In chapter 3 (Patterns of Inheritance) read the section for Triplet Repeat Expansion Disorders on pages 54 – 57 (See also Figure 3.28 which is on page 58).
 - b. Chapter 3, see Hot Topic 3.1 on 56 and 57 – for FXTAS – which describes how, in the case of fragile X, the premutation allele has a disorder associated with **overexpression** of the *FMR1* gene, while fragile X syndrome itself is caused by **loss** of expression of *FMR1*.
 - c. Chapter 4 (The Human Genome) see Clinical Snapshot 4.1 – Huntington disease on page 74
 - d. Chapter 16 (Pharmacogenetics) on pages 230 – 237. Note that this is a very short chapter which focuses on the case of a boy who develops malignant hyperthermia during tonsillectomy surgery.

Videos: As posted on Panopto site

1. Triplet repeat expansion disorders which includes
 - General features of unstable triplet repeat expansions
 - Fragile X syndrome
 - Huntington disease
 - Myotonic dystrophy, Friedreich ataxia and the SCA disorders
2. Pharmacogenetics
 - pharmacogenetic concepts
 - Malignant hyperthermia
 - CYP2D6, Metoprolol, and Codeine